

IT3091 – Bank Marketing

Preprocessing Log

#	Step	Decision Made	Rationale / Evidence	Result
1	Missing value handling — job/education/contact	Filled with explicit "unknown" category	Genuine unrecorded values (EDA log row 2): job 288 (0.64%), education 1,857 (4.11%), contact 13,020 (28.80%). Too large a share (esp. contact) to drop or impute confidently.	0 nulls remaining in all three columns
2	Missing value handling — poutcome	Filled with "no_prior_campaign" (not "unknown")	Missingness is structural, not random: null in 100% of rows where pdays = -1, only 0.06% otherwise (EDA log rows 2 & 7). Represents "not applicable," not "unrecorded."	0 nulls remaining; 36,959 rows recoded
3	Feature engineering — contact history flag	Created binary previously_contacted = 1 if pdays != -1 else 0	Makes the sentinel-driven signal in pdays explicit and usable by any model type (EDA log row 5).	Mean 0.1826 → confirms 18.26% of clients previously contacted, matching earlier EDA figure
4	Feature engineering — pdays sentinel	Attempted: replaced -1 with max + 1 (872) as pdays_capped. Reversed after preprocessing testing.	RobustScaler produced a broken scale (-871 to 0) because >75% of values shared the sentinel replacement, making IQR = 0. Decided to drop pdays_capped entirely rather than force a fix, since previously_contacted + previous already capture the same signal (r = 0.45, EDA log row 9) without redundancy.	pdays_capped and raw pdays both excluded from final feature set

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5	Leakage removal	Dropped duration from the modeling dataset; retained a separate df_diagnostic copy that includes it	Median call duration 164s (no) vs 426s (yes) — ~2.6x difference confirms leakage (EDA log row 6); only known after a call takes place	duration excluded from bank_train.csv/bank_test.csv; available separately for an optional diagnostic-only benchmark model
6	Column-name verification	Confirmed the day-of-month field is named day_of_week in this data pull (not day), despite representing day of month (1–31), not weekday	Verified directly against the loaded dataframe's column list before scaling	Numeric scaling list corrected to use day_of_week
7	Categorical encoding	One-hot encoded job, marital, education, default, housing, loan, contact, month, poutcome with drop_first=True	Standard encoding for nominal categories; month flagged as cyclical — one-hot used as baseline, cyclical (sin/cos) encoding noted as a possible refinement for model comparison stage	Feature set expanded from 17 to 43 columns (incl. target)
8	Target encoding	Mapped y: "no" → 0, "yes" → 1	Required numeric target for classification models	Success
9	Train/test split	Stratified 80/20 split, random_state=42	Class imbalance (88.3% / 11.7%) risks an uneven split without stratification (EDA log row 3)	Train: 36,168 rows; Test: 9,043 rows. Class ratio preserved at 88.30%/11.70% in both sets
10	Feature scaling	RobustScaler (median/IQR-based) fit on training set only, applied to age, balance, day_of_week, campaign, previous	Chosen over StandardScaler/MinMaxScaler because of documented outliers and skew (EDA log row 4); fit on train only to prevent test-set leakage into the scaler	Scaled features centered near 0 with skew-resistant spread

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11	Data versioning & storage	Saved bank_cleaned.csv (pre-encoding, human-readable), bank_train.csv, bank_test.csv to Data/PreprocessedData in the team GitHub repo	Gives every team member (modeling, evaluation, recommendation) the identical, reproducible train/test split needed for fair model comparison	Files committed and pushed; accessible via git pull